

Charbel Junior Nour

(647) 949-5701 | linkedin.com/in/cjnour | cjnour.com | charbeljrour@gmail.com

PROFESSIONAL SUMMARY

Motivated Electrical Engineering new graduate from McMaster University with 2 years of professional experience at Tesla and Moneris. Passionate about controls engineering and automation, with a proven ability to deliver innovative solutions – ranging from industrial projects in autonomous systems to software-driven IoT solutions. Recognized for developing impactful, standardized designs and excelling in collaboration in cross-functional environments. Known for creativity, adaptability, and a results-driven mindset, I bring a blend of rich practical experience and fresh academic knowledge to solve complex challenges and drive meaningful outcomes.

CORE COMPETENCIES

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|---------------------|----------|-------------------------------|
| • Control Systems | • C++ | • Object-Oriented Programming |
| • PLC Programming | • Python | • Microprocessors |
| • Beckhoff TwinCAT3 | • Java | • Computer Aided Design (CAD) |

PROFESSIONAL EXPERIENCE

Tesla

Controls Engineer Intern

Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and developed a **custom HMI dashboard** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Contributed to commissioning **3 automation** machines by leveraging **Beckhoff TwinCAT3 PLC** and **Safety PLC logic** to program critical components, including **VFDs**, **IPCs**, sensors, and robotic and pneumatic devices.
- Led multiple **R&D** initiatives with an aim to **solve critical engineering challenges** to enhance machine performance in production environments by experimenting with **time-of-flight lasers** and **profilometers**.

Material Planner Intern

Sep. 2023 – Dec. 2023

- Developed and maintained 3 internal tools that streamlines the part reordering process through analysis of kitting history, conducting comprehensive **gap analyses**, and employing advanced **Excel VBA** techniques with existing company inventory – resulting in a significantly efficient database.
- Utilized these tools to build a software analysis team that focuses on the automation and standardization of many daily processes, resulting in a **reduction of duplicate shipments by 100%**.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

Moneris

Technology Governance Intern

May 2023 – Aug. 2023

- Revamped the internal application database and cut its size **by over 50%** through **cost center analysis** and a detailed **general ledger audit**. Coordinated the retirement of outdated applications and consolidated underutilized software licenses, resulting in **significant cost savings**.
- Orchestrated collaboration across departments to gather, validate, and standardize data for **cloud migration planning**, ensuring a seamless transition phase and enhancing application scalability.
- Strengthened IT governance and compliance frameworks by aligning the application inventory with corporate standards to streamline software usage across teams, resulting in a reduction of licensing costs.

Software Engineer Intern

Jun. 2022 - Apr. 2023

- Developed and sustained an internal tool that increases productivity for **30+ employees**. The tool displays VISA, MasterCard, and AMEX transaction analytics by using **JavaScript** and **React** hooks to build a responsive and compatible UI that can manage dashboards, widgets, and layout customization.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted by **Parse** to reduce boilerplate, implement reducers, utilize dispatch, and **query user database** properties.
- Collaborated with the QA team in semi-weekly **agile** feedback sessions to identify and address platform issues, enhancing the reliability of a proprietary testing tool for payment platform evaluations.

EXTRACURRICULARS

McMaster EcoCar EV Challenge

Propulsion Controls and Modeling Team Member

Sep. 2024 - Present

- **Modelling control systems** in **Simulink** for a Cadillac Lyriq EV through the use of documentation provided by **General Motors**, ensuring precise alignment with vehicle specifications and our team's performance goals.
- **Testing and validating control system requirements** using **Simulink**, ensuring compliance with benchmarks and functionality standards outlined by the competition organizers to achieve an **enhanced propulsion system**.
- Developing **energy-efficient** and **customer-focused control features** as part of a competition sponsored by the **US Department of Energy** and **General Motors** involving 13 universities across North America.

PROJECTS

Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection** and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

Autonomous RC Car

- Implemented self-driving algorithms coded in **C++** and **Python** on an RC car to navigate through obstacles via **LiDAR** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **semi-weekly** agile team meetings to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

Smart Curtain System

- Created a solution for sleep deprivation by designing an automated curtain. The app was coded in **React Native** and the backend is hosted on **AWS IoT** to store data for alarm functions and automated sunrise/sunset control.
- Implemented and deployed a **C++** program on an **Arduino Uno** to contact the backend server and control the stepper motor, enabling precise movement for rolling the curtain up and down automatically.

Wireless Dog Doorbell System

- Designed and developed an **affordable wireless doorbell system** with an **Arduino Uno**, enabling dogs to alert owners when they need to go outside, facilitating effective potty training through dog-to-human communication.
- Built a mobile app using **React Native** to send real-time push notifications to owners, ultimately enhancing communication and improving the effectiveness of the training process.

Automated Vanity System

- Designed and fabricated a custom automated vanity system by 3D-modeling a recycled oil barrel in **Autodesk Inventor**. Cut and welded a panel from the barrel to a recycled chair and added wheels for smooth accessibility, and ensured the chair sat flush when retracted.
- Engineered a motorized mirror and a Bluetooth sound system by wiring a recycled car window regulator motor to a rocker switch and a power supply with an **AC to DC rectifier**. Researched **torque-speed characteristics** and motor specifications to ensure the motor could support the load with enough power from the power supply.

EDUCATION

B. Eng., Electrical Engineering

McMaster University

Expected Apr. 2025

Hamilton, ON

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, High Performance Programming, Principles of Programming, Microprocessor Systems, Signals & Systems, Logic Design, Control Systems, Intro to Mechatronics, Power Electronics, Electric Motor Drives, Electromagnetics I & II, Electric Devices & Circuits
- **Extracurriculars:** McMaster EcoCar EV Challenge PCM Team Member, GDSC 2023 Incubator Challenge
- **Special Interests:** Guitar, Music Production, 3D Printing, Basketball, Skiing, Cycling, Running